

Calibration Sheet



Instrument: CONTROS HydroC CO₂ FT Date of calibration: 08.01.2025
Serial number: CO2FT-0918-001
Customer: SMHI PO number: RMA71121

Sensor characteristics

Note! For proper handling please check our manual, esp. chapter "Important Notices"!

Energy consumption

CONTROS HydroC™ FT standard, general values:

Current	20 °C	During warm-up**
@ 12 V	440 mA	710 mA
@ 24 V	280 mA	1060 mA

* measured during calibration

** warm-up time decreases with increasing voltage

[consumption during zeroing: +330 mA @ 12 V]

Warm-up time

(T_{control}, general values of standard CONTROS HydroC™ FT):

Warm-up time at 7 °C water temperature		Voltage	
		12 V	24 V
Ambient temperature	14 °C	31 min	
	20 °C	19 min	3 min
	25 °C	16 min	

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Sensor specifications

Operating depth:	Surface, flow through sensor
Gas tension (abs.):	800 mbar to 1100 mbar (defined through measuring range of the internal pressure sensors p_NDIR and p_in)
Measuring range:	200 µatm to 1000 µatm
Resolution – digital signal:	<1 µatm
Initial accuracy:	± 0.5 % reading (as total sum of all error)
Water temperature range:	1 °C to 30 °C
Calibration optimised for:	1 °C to 30 °C (<i>as requested by customer</i>)
Ambient temperature:	1 °C to 35 °C (The instrument should not be operated at ambient temperatures lower than 8 °C compared to the water to avoid condensation within the gas stream.)
Dimensions:	325 x 240 x 136 mm (L x W x H)
Weight:	5.3 kg
Output connector:	Hirschmann-plug CA6LD
Voltage:	11 to 24 V
RS232/ EIA-232 output signal:	See “CONTROS HydroC® On-line data”
RS485/ EIA-485 output signal:	See “CONTROS HydroC® On-line data”
Baud rate (preset):	115200 Bd
Interval settings (preset):	Warm-up: 0 min; Zero: 2 min; Flush: 0 min; Measure: 718 min
Data logger:	2 GB

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Calibration details

Setup and parameters

Calibration unit:	Contros CO ₂ calibration tank #1
RS232 protocol unit:	DETECT-2 software package on CO-CALIB2
Voltage:	24.0 VDC
Temperature:	14.0 °C
Ambient temperature:	22.0 °C
Water:	Deionised water with carbonate additives
Salinity:	0 ‰
Reference gases:	Zero Air, 398.3 ppm 801.8 ppm CO ₂ in natural air
Reference system:	SPRINK underway instrument with LiCOR LI7815
Pump Flow rate:	approx. 6.5 L/min
Pressure in water stream:	+100 mbar with respect to ambient pressure

Sensor

Membrane:	TOUGH membrane
Control temperature (NDIR detector):	50 °C

Data Processing Sheet



Instrument: HydroC CO2
Serial number: CO2FT-0918-001
Customer: SMHI

Date of calibration: 08.01.2025 (pre 14°C)
PO: RMA71121

Note! *For more information about the HydroC calibration, please check your individual sensor Calibration Sheet.*

Note! *For data processing, apply the application note Data Processing for CONTROS HydroC CO₂.*

Sensor Specific Values

T_0	273.15 K
p_0	1013.25 mbar
F	61470
T_{sensor}	47.9°C
$f(T_{\text{sensor}})$	9849.19 (only for T_{sensor} as given above)
$S'_{2\text{beam},Z}$	14471.99 (found during calibration)
Polynomial degree	3 (with forced zero crossing)
Regression error:	<0.6 ppm (estimate error found during calibration)
Runtime:	137046260 s

Calibration coefficients

k_1	5.124807e-02
k_2	5.222492e-06
k_3	6.372735e-11

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Calibration Data

S_{raw}	S_{ref}	T_{gas}	p_{NDIR}	S_{proc}	$x_{\text{CO}_2, \text{reference}}^*$
[]	[]	[°C]	[mbar]	[]	[ppm]
20037.94	15132.80	40.56	1005.52	6075.18	599.19
19516.67	15138.42	40.45	1004.66	7536.29	822.44
20533.82	15128.83	40.40	1003.63	4689.42	420.04
21269.82	15118.42	40.37	1003.12	2613.73	197.64

Equations

Equation for $x_{\text{CO}_2, \text{wet}}$

$$x_{\text{CO}_2, \text{wet}} = (k_3 S_{\text{proc}}^3 + k_2 S_{\text{proc}}^2 + k_1 S_{\text{proc}}) \frac{p_0 T_{\text{gas}}}{T_0 p_{\text{NDIR}}}$$

Equation for p_{CO_2}

$$p_{\text{CO}_2} = x_{\text{CO}_2, \text{wet}} \frac{p_{\text{in}}}{1013.25}$$

Calibration Curve

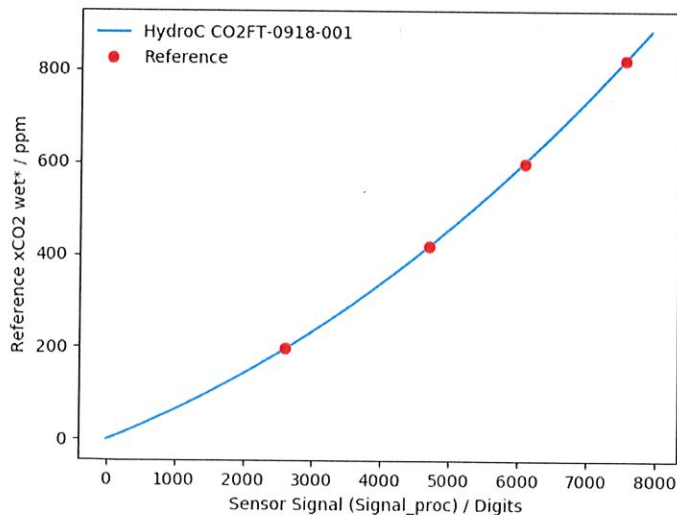


Figure 1: Calibration curve of the processed sensor signal (S_{proc}) against the x_{CO_2} of the Contros CO₂ reference system.

*Converted from the x_{CO_2} value in the reference system to the conditions in the gas stream of the sensor.

Data Processing Sheet



Instrument: HydroC CO2
Serial number: CO2FT-0918-001
Customer: SMHI

Date of calibration: 10.12.2024 (post 14°C)
PO: RMA71121

Note! *For more information about the HydroC calibration, please check your individual sensor Calibration Sheet.*

Note! *For data processing, apply the application note Data Processing for CONTROS HydroC CO₂.*

Sensor Specific Values

T_0	273.15 K
p_0	1013.25 mbar
F	61470
T_{sensor}	47.9°C
$f(T_{\text{sensor}})$	9849.18 (only for T_{sensor} as given above)
$S'_{2\text{beam},Z}$	14474.72 (found during calibration)
Polynomial degree	3 (with forced zero crossing)
Regression error:	<0.6 ppm (estimate error found during calibration)
Runtime:	136777692 s

Calibration coefficients

k_1	5.209697e-02
k_2	4.421106e-06
k_3	1.236227e-10

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Calibration Data

S_{raw}	S_{ref}	T_{gas}	p_{NDIR}	S_{proc}	$x_{\text{CO}_2, \text{reference}}^*$
[]	[]	[°C]	[mbar]	[]	[ppm]
19507.59	15131.83	40.92	1044.06	7548.04	779.44
19973.13	15126.83	40.51	1043.38	6242.96	587.66
20557.67	15120.02	40.46	1042.57	4601.13	386.16
21217.81	15109.69	40.49	1041.99	2734.83	198.43

Equations

Equation for $x_{\text{CO}_2, \text{wet}}$

$$x_{\text{CO}_2, \text{wet}} = (k_3 S_{\text{proc}}^3 + k_2 S_{\text{proc}}^2 + k_1 S_{\text{proc}}) \frac{p_0 T_{\text{gas}}}{T_0 p_{\text{NDIR}}}$$

Equation for p_{CO_2}

$$p_{\text{CO}_2} = x_{\text{CO}_2, \text{wet}} \frac{p_{\text{in}}}{1013.25}$$

Calibration Curve

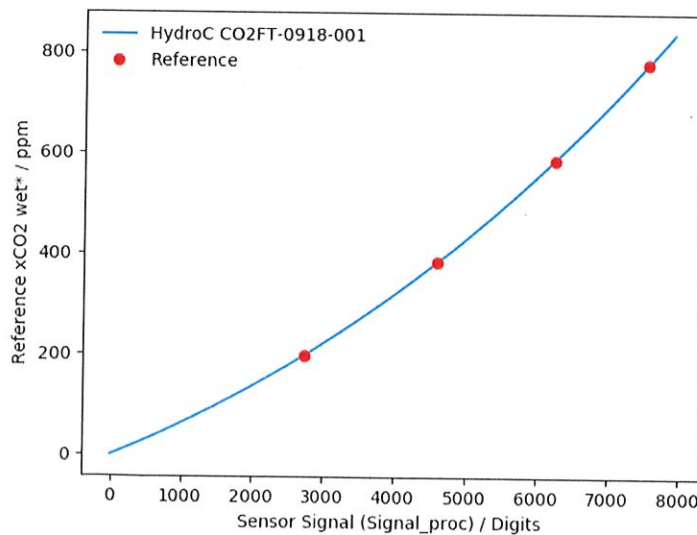


Figure 1: Calibration curve of the processed sensor signal (S_{proc}) against the x_{CO_2} of the Contros CO₂ reference system.

*Converted from the x_{CO_2} value in the reference system to the conditions in the gas stream of the sensor.